INCH-POUND
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PERFORMANCE SPECIFICATION

CHASSIS, TANK: ARMORED VEHICLE, BRIDGE LAUNCHER, M60A1

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of full-tracked tank chassis which with subsequent adaption mounts a 60 foot (ft) (18 meter (m)) launched bridge. This specification is limited to the vehicular chassis of the launcher.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2350

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Superintendent of Documents, U.S. Government Printing Office, Washington,
DC 20402.)

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DRAWING

ARMY

10934200 - Tank, Chassis, M60A1, Transporter, AVL Bridge.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted should be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes (DoD Adopted).

(Copies of ANSI documents are available from the American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document should takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless specified herein, materials shall be in accordance with the manufacturer's materials specifications. The materials shall be capable of meeting all the operational and environmental requirements specified herein. All rubber used in construction of the chassis shall be ozone resistant. Asbestos, cadmium and radioactive materials shall not be used in this item. Radioactive material is defined by Title 10, CFR, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.5.1).

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3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.5.1).

3.3 Design and construction. Chassis components, subassemblies, and assemblies shall be fabricated into a complete chassis in accordance with Drawing 10934200, drawings subordinate to 10934200, and applicable specifications and standards. All parts requiring identification shall be marked in accordance with the drawing (see 4.5.2).

3.3.1 Chassis reliability. Chassis shall demonstrate a minimum of 180 hours Mean-Time-Between-Failure (MTBF) during 4000 miles (mi) (6400 kilometers (km)) of operational tests in accordance with the operational profile (see 3.3.3). For calculating MTBF, the definition of combat mission failures in the M60A1 Tank RAM-D failure definition and scoring criteria shall be used (see 4.2.2.3).

3.3.2 Chassis maintainability. The chassis shall not exceed the following maintenance ratios (see 4.2.2.4):

Organizational maintenance:	0.05
Direct support maintenance:	0.04
Combined maintenance:	0.09

The maintenance ratio shall be calculated by dividing the total maintenance man-hours (exclusive of daily crew checks and services) by the total operating hours.

3.3.3 Operational profile. Profile shall consist of (see 4.2.2.2):

a. 25 percent (%) on paved roads, either concrete or asphalt or any combination of the two.
b. 25% on gravel and dirt roads with at least 10% of this distance under mud conditions.
c. 25% on level, cross country.
d. 25% on hilly, cross country.

3.4 Performance. After being subjected to the break-in run and procedures of 6.6, the chassis, loaded to simulate an M60A1 vehicle weight of 112 600 pounds (lbs) (51 200 kilograms (kg)), shall perform as specified herein under all operating conditions (see 4.5.4).

3.4.1 Power plant and power train. The power plant shall function throughout all gear and speed ranges without loss of lubricants or damage that may cause failure of the chassis. The power train and associated controls shall be capable of operation throughout all speed and

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steering ranges without binding of linkages, loss or lubricants, grabbing, chattering or slippage when controls are applied to steer, stop, or hold chassis (see 4.5.4.1).

3.4.2 Environments. The chassis shall operate in ambient air temperature of 115 degrees Fahrenheit (°F) to minus (-) 25°F (46 degrees Celsius (°C) to -31°C) and with special equipment installed, at temperatures to -65°F (-53°C) (see 4.5.4.2).

3.4.2.1 Cooling system. With chassis operating in ambient temperature of 115°F (46°C), the engine oil temperature shall not exceed 250°F (121°C) at the heat exchanger outlet. The transmission oil temperature shall not exceed 300°F (150°C) at the heat exchanger inlet. Temperature indicators on the driver's panel shall remain in the green area (see 4.5.4.2.1).

3.4.3 Level road acceleration. The chassis shall accelerate from a standing start on level road through a distance of 200 ft (61 m) in a time period of not more than 13 seconds (see 4.5.4.3).

3.4.4 Level road speeds. The chassis shall operate at sustained minimum and maximum speeds of 2.5 miles per hour (mph) (4 kilometers per hour (km/h)) and 30 mph (48 km/h) on level road without damage to power plant and power train (see 4.5.4.4).

3.4.5 Level road drift. When traveling between 25 and 30 mph (40 and 48 km/h) on level road, in a straight line the chassis drift shall not exceed 3 ft (1 m) in 100 ft (30 m), without manual steer correction (see 4.5.4.5).

3.4.6 Level road stopping. The chassis traveling at 20 mph (32 km/h) on level road, shall stop within 60 ft (18 m) from point of brake application. Drift shall not exceed 4 ft (1.2 m). Steering correction is allowed (see 4.5.4.6).

3.4.7 Grades and slopes. Chassis shall ascend longitudinal grades of 60% in forward and reverse gear without stalling or damage to power plant and power train. Chassis shall operate on right and left side slopes of 30%. During grade and slope operations, normal engine lubricating oil pressures shall be maintained (see 4.5.4.7).

3.4.7.1 Engine starting (grades and slopes). Chassis standing on a 60% in grade for not less than 2 minutes and with engine idling under no load between 700 and 750 revolution per minute (rpm), the engine shall be stopped for not less than 2 minutes. The engine shall restart in not more than 1 minute when headed either up or down grade. A similar engine stopping and starting test shall be accomplished on 30% right and left side slopes (see 4.5.4.7).

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3.4.7.2 Fuel supply (grades and slopes). During engine operation, the fuel system shall maintain fuel supply to the engine when ascending and descending 60% grades in forward and reverse gear, and when chassis is being operated on 30% side slopes with either side of chassis up slope (see 4.5.4.7).

3.4.7.3 Holding. With the chassis standing on a 60% grade, with service brakes applied, the chassis shall be held stationary when headed either up or down grade. With parking brake engaged and all other holding devices inoperative, the chassis shall be held stationary when headed either up or down grade (see 4.5.4.7).

3.4.7.4 Grade speeds. The chassis shall operate at a sustained speed of 10 mph (16 km/h) while ascending a 10% grade and at 20 mph (32 km/h) while ascending a 3% grade (see 4.5.4.7).

3.4.8 Shallow water fording. Chassis shall ford a level, hard-bottom body of water 48 inch (in.) (1220 millimeter (mm)) in depth, including wave, without special equipment. With the chassis standing in water 48 in. (1220 mm) in depth for 30 minutes, the accumulation of water shall be not more than 1.5 in. on the crew compartment hull floor measured in the center of the "V" (see 4.5.4.8).

3.4.8.1 Engine starting (shallow water). With the chassis standing for 30 minutes in water 48 in. (1220 mm) in depth, the engine operating at 1000 rpm for 15 minutes, then stopped for 15 minutes, the engine shall restart in not more than 3 minutes. All accessories shall function satisfactorily during and after fording operations (see 4.5.4.8.1).

3.4.8.2 Lubricant contamination. After fording operations, the water content of the transmission, engine, final drives, and suspension system lubricants shall be not more than 2% by volume (see 4.5.4.8.2).

3.4.9 Trench crossing. The chassis shall cross trenches 36 in. (915 mm) in depth and 102 in. (2590 mm) in width without stalling or damage to suspension and other vehicle equipment (see 4.5.4.9).

3.4.10 Vertical obstacles. The chassis shall cross over vertical obstacles 36 in. (915 mm) in height while moving forward without stalling or damage to suspension and hull floor, speed not to exceed 5 mph (8 km/h) (see 4.5.4.10).

3.4.11 Turning. The chassis shall turn 360° to the right and left in pivot, in neutral steer, within a circle 35 ft (10.7 m) in diameter (see 4.5.4.11).

3.4.12 Sealing. Except as specified in 3.4.8, all seals and sealer shall prevent the entrance of water and foreign matter under all operating conditions. Static seals shall provide an interference fit of sealing surfaces preventing the leakage of fluids intended to be contained by the

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seals. Dynamic seals shall prevent fluid leakage in the form of a drop (see 6.5) when the chassis is standing idle. Evidence of lubricant at seals specifically designed to allow exit of flushed lubricants shall be permitted (see 4.5.4.12).

3.4.12.1 Escape hatch seal. The escape hatch seal shall prevent entrance of water into the crew compartment, under all operating conditions (see 4.5.4.12.1).

3.4.13 Controls. When installed in accordance with applicable drawings, all electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all chassis operating conditions (see 4.5.4.13).

3.4.14 Adjustment mechanisms. When installed and adjusted in accordance with applicable drawings, all adjustment mechanisms shall function properly and maintain adjustment settings during all chassis operating conditions (see 4.5.4.14).

3.4.15 Generator.

3.4.15.1 Generator voltage. Generator voltage shall be regulated to not less than 27.8 Vdc and not more than 28.3 Vdc (see 4.5.4.15).

3.4.15.2 Generator blower motor. The generator blower motor shall be capable of delivering a constant flow of air through the generator (see 4.5.4.15).

3.4.16 Air cleaner blower motors. Each air cleaner blower motor shall provide a flow of air through each blower outlet. The air cleaner blower motors shall be wired to prevent blower operation when engine is not running (see 4.5.4.16).

3.4.17 Lights. All lights shall operate as specified on applicable drawings through all chassis operating conditions. All electrical contacts and connections shall maintain positive contact under all chassis operating conditions (see 4.5.4.17).

3.4.18 Locking, unlocking and actuated mechanisms.

3.4.18.1 Hull drain valves. After unlocking lever, the force required to operate the front drain valve lever shall not exceed 17 lbs (7.7 kg) at the end of the hand grip to actuate the valve a minimum of 0.375 in. (9 mm) travel. After unlocking lever, the force required to operate the rear drain shall not exceed 25 lbs (11.5 kg) at a point just below the knob to actuate the valve through a minimum of 0.687 in. (17 mm) travel (see 4.5.4.18.1).

3.4.18.2 Escape hatch release. The force required to operate the handle to release the escape hatch shall be 50 $\pm$10 lbs (23 $\pm$ 5 kg) pull at a point 0.5 in. (13 mm) from the end of the handle (see 4.5.4.18.2).

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3.4.18.3 Operator and assistant operator's seat. With the operator or assistant operator in seated position, the force required to actuate the fore and aft seat adjustment lever shall not exceed 12 lbs (5.5 kg) at the end of the lever (see 4.5.4.18.3).

3.4.18.4 Fuel shutoff valve. Engine shall stop firing and rotating within a maximum of 30 seconds after the manual fuel shutoff valve is actuated to the "OFF" position (see 4.5.4.18.4).

3.4.19 Fire extinguisher. The fixed fire extinguisher system, when installed in accordance with applicable drawings, shall be readily accessible for operation internally or externally. Actuating the internal controls shall simultaneously stop the supply of fuel to the engine. The maximum effort required to discharge the system either internally or externally shall be not more than 55 lbs (25 kg). A time delay of more than 6 seconds but less than 11 seconds shall occur on the first shot between actuating of the release mechanism and entrance of CO_2 into the engine compartment (see 4.5.4.19).

3.4.20 Personnel heater. The heater shall be capable of starting with the switch in either high or low position. The blower motor shall operate at reduced speed when the heater is turned on. The burner shall ignite within 4 minutes of heater turn-on. Subsequent to ignition, the heater shall be capable of generating high or low heat and the blower motor shall run at full speed. Turning the heater "off" shall cause both the burner to extinguish and the blower motor to shut off within 3.5 minutes (see 4.5.4.20).

3.4.20.1 Heater fuel pressure. With either of the fuel system pumps and the heater pump operating, the chassis fuel system shall produce a minimum pressure of 7 pound per square inch (psi) (48 kilopascals (kPa)) at the outlet of the heater pump line (see 4.5.4.20).

3.4.21 Engine smoke generator. The chassis shall discharge smoke from both exhaust pipes when the engine is running and the smoke generator switch is in the "on" position (see 4.5.4.21).

3.4.21.1 Smoke grenade discharger circuits.

3.4.21.1.1 Arming. The power-on (ready) lamp in the grenade power box shall illuminate when the master battery switch and the grenade system power switch are both ON (see 4.5.4.21.1.1).

3.4.21.1.2 Circuit voltage. Voltage at the smoke grenade discharger firing pins shall be indicated at table I when the power-on (ready) lamp in the grenade power box is illuminated and the appropriate firing pushbutton is actuated (see 4.5.4.21.1.2).

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TABLE I. Smoke grenade discharger firing pin voltage.

Firing pushbutton	Left side discharger firing pins	Right side discharger firing pins	Voltage
Right	1, 2, 5	3, 4, 6	Vehicle (ref. 3.6)
Right	3, 4, 6	1, 2, 5	0 ± 0.5 Vdc
Left	3, 4, 6	1, 2, 5	Vehicle (ref. 3.6)
Left	1, 2, 5	3, 4, 6	0 ± 0.5 Vdc

3.4.22 Vibration and noise level. Equipment operation shall not exceed the maximum acceptable noise level specified in table II (see 4.5.4.22).

TABLE II. Maximum acceptable noise levels for Army material command equipment (continuous noise).

Octave band center frequency	Maximum acceptable noise level (dB ref. 0.0002 microbar)
63	121
125	111
250	103
500	102
1000	100
2000	100
4000	100
8000	100

3.5 Stowed equipment. The chassis shall be capable of stowing all on-vehicle equipment (OVE) items in the spaces provided (see 4.5.2).

3.6 Electrical system. The chassis electrical system shall consist of the hull electrical subsystem as described below. System shall be 24 volts direct current (Vdc), nominal (see 4.5.2).

3.6.1 Power plant electrical. The power plant electrical equipment shall consist of the following: sending units necessary to provide engine data in the form of electrical signals to the operator's instrument panel, an engine electrical disconnect to facilitate engine removal and replacement, an electrical starter motor capable of starting the engine, and an air-cooled generator (see 4.5.2).

3.6.2 Interior lighting. Each crew position shall contain a domelight to provide general illumination when required. The domelight shall contain provisions for adjusting light intensity and red filter for night vision adaption (see 4.5.2).

3.6.3 Auxiliary outlet. One outlet (24 Vdc nominal) shall be provided in the hull to mate with the OVE trouble light assembly. The auxiliary outlet circuit shall be protected with a

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15 ampere, automatic reset, circuit breaker (see 4.5.2).

3.6.4 Engine manifold heater. The wiring for the engine manifold heater shall provide a switch-actuated chassis battery circuit to the engine. Operation of the pushbutton switch on the purge pump handle, with the starter switch depressed, shall provide chassis battery voltage to pin "a" of the multiple connector at the engine electrical quick disconnect (see 4.5.2).

3.6.5 Communications system power supply. Provisions shall be made through a mating connector to supply power to the communications system from the hull electrical system (see 4.5.2).

3.7 Fluid systems.

3.7.1 Hydraulic system. Provisions shall be made to assure internal cleanliness of the chassis hydraulic system. Hydraulic lines and connections shall be free from leakage in excess of a seep (see 6.5).

3.7.2 Fuel system. Provisions shall be made to assure internal cleanliness of the fuel system prior to initial fueling. Fuel system shall be free from leaks (see 4.5.2).

3.7.3 Fuel tanks (rapid fill). Fuel tanks shall be capable of receiving 50 gallons (gal) (190 liters (L)) of fuel per minute (see 4.5.2).

3.8 Operator's protection. Noise hazard caution signs shall be posted in all crew stations. The signs shall be clearly visible to all personnel (see 4.5.2).

3.9 Painting, marking, and data plates.

3.9.1 Painting. Unless otherwise specified (see 6.2), chassis exterior paint shall be forest green camouflage, lusterless, alkyd enamel conforming to MIL-C-46168. The engine compartment shall be gloss white, matching color chip 17875 of FED-STD-595. The crew compartment shall be semigloss green, matching color chip 24533 of FED-STD-595. All sliding parts and operating contact surfaces shall be free of paint (see 4.5.2).

3.9.2 Marking. Marking paint shall be lusterless black, matching color chip 37038 of FED-STD-595 (see 4.5.2).

3.9.3 Identification and data plates. All plates shall be in accordance with applicable drawings (see 4.5.2).

3.10 Workmanship. All parts, components and assemblies of the chassis shall be clean and free from sand, dirt, fins, pits, cracks, sprues, scales, flux and other harmful extraneous materials and defects. External surfaces shall be free from burrs, sharp edges and corners except

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when sharp edges and corners are required. Electric wires and air lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. First article inspection shall be performed on preproduction or initial production samples when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table III) and the tests of 4.5.4 through 4.5.4.22 (see table IV).

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction inspection shall be performed on one sample. This inspection shall include the examinations of 4.4 (see table III) and the tests of 4.5.4 through 4.5.4.22 (see table IV).

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article preproduction inspection shall be performed on one sample. This examination shall include the examinations of 4.4 (see table III) and the test of 4.5.4 through 4.5.4.22 (see table IV). Two additional chassis from the first month production or the first 20 produced shall be subjected to the initial production test.

4.2.2.1 Initial production test. To determine conformance to section 3 (inclusive), vehicles will be examined as specified in 4.5.4 and tested as specified in table V. Subsequently, the vehicles shall be tested as specified in 4.2.2.2. Tests shall be performed by the Government at a site approved by the Government and shall require no more than 120 days. Delays caused by vehicle breakdown due to poor quality of workmanship or material or failure of the contractor to comply with specification or drawing requirements, shall not be the basis for adjustment of the contract performance dates, delivery schedule, or contract price. The contractor shall expeditiously furnish repair parts as required to support testing.

4.2.2.2 Test profile. Second test chassis shall be subjected to four repetitions of the 1000 mi (1600 km) cycle contained in table V for a total test mileage of 4000 mi (6400 km). Third test chassis shall be subjected to the initial production tests and then to four repetitions of 1000 mi (1600 km) cycle contained in table V.

4.2.2.3 Reliability verification. To determine conformance to 3.3.1, a point estimate of the MTBF equal to greater than 180 hours must be demonstrated during the test.

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4.2.2.4 Maintainability verification. To determine conformance to 3.3.2, it shall be determined, during test, that point estimates of maintenance ratios are not exceeded.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1, 4.5.4.2 and 4.5.4.3 (see table IV). Sampling shall be in accordance with 4.5.3.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Materials - Not as specified herein and in drawings (see 3.2).	Visual
102	Fabrication - not in accordance with specified drawings, standards and specifications (see 3.3).	Visual & SIE 3/
103	Reliability - MTBF less than specified (see 3.3.1).	Visual
104	Maintainability - Maintenance ratios exceed specified maximums (see 3.3.2)	Visual
105	Operational profile - profile used to determine MTBF not as specified (see 3.3.3).	Visual
106	Stowed equipment - insufficient stowage space (see 3.5).	Visual
107	Electrical system - voltage not as specified (see 3.6).	Visual & SIE
108	Power plant electrical - voltage not as specified (see 3.6.1).	SIE
109	Engine manifold heater - wiring not as specified (see 3.6.4).	Visual
110	Communications system power supply - power not supplied from hull electrical system (see 3.6.5).	Visual
111	Hydraulic system - leakage exceeding a seep (see 3.7.1).	Visual
112	Fuel system - leakage (see 3.7.2). 1/	Visual
113	Fuel tanks - insufficient filling capacity (see 3.7.3).	SIE
114	Operator's protection - noise hazards not posted (see 3.8).	Visual
115	Painting - Not as specified (see 3.9.1).	Visual
116	Workmanship - Insufficient (see 3.10).	Visual
Minor:		
201	Interior lighting - domelight or red filter not provided, intensity not adjustable (see 3.6.2). 2/	Visual
202	Auxiliary outlet - not provided, not protected (see 3.6.3).	Visual
203	Marking - Not as specified (see 3.9.2).	Visual
204	Identification and data plates -Not in accordance with drawings (see 3.9.3)	Visual

1/ Fuel leaks constitute a special defect and will result in total inspection of the lot for this defect.
2/ Filament failures are not to be classified as major or minor defects.
3/ SIE = Standard Inspection Equipment.

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TABLE IV. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance
Power plant & power train	3.4.1	4.5.4.1	X	X
Environments	3.4.2	4.5.4.2	X	X
Cooling system	3.4.2.1	4.5.4.2.1	X	
Level road acceleration	3.4.3	4.5.4.3	X	X
Level road speeds	3.4.4	4.5.4.4	X	
Level road drift	3.4.5	4.5.4.5	X	
Level road stopping	3.4.6	4.5.4.6	X	
Grades & slopes	3.4.7	4.5.4.7	X	
Engine starting	3.4.7.1	4.5.4.7	X	
Fuel supply	3.4.7.2	4.5.4.7	X	
Holding	3.4.7.3	4.5.4.7	X	
Grade speeds	3.4.7.4	4.5.4.7	X	
Shallow water fording	3.4.8	4.5.4.8	X	
Engine starting	3.4.8.1	4.5.4.8.1	X	
Lubrication contamination	3.4.8.2	4.5.4.8.2	X	
Trench crossing	3.4.9	4.5.4.9	X	
Vertical obstacles	3.4.10	4.5.4.10	X	
Turning	3.4.11	4.5.4.11	X	
Sealing	3.4.12	4.5.4.12	X	
Escape hatch seal	3.4.12.1	4.5.4.12.1	X	
Controls	3.4.13	4.5.4.13	X	
Adjustment mechanisms	3.4.14	4.5.4.14	X	
Generator voltage	3.4.15.1	4.5.4.15	X	
Generator blower motor	3.4.15.2	4.5.4.15	X	
Air cleaner blower motor	3.4.16	4.5.4.16	X	
Lights	3.4.17	4.5.4.17	X	
Hull drain valves	3.4.18.1	4.5.4.18.1	X	
Escape hatch release	3.4.18.2	4.5.4.18.2	X	
Operator and assistant operator's seat	3.4.18.3	4.5.4.18.3	X	
Fuel shutoff valve	3.4.18.4	4.5.4.18.4	X	
Fire extinguisher	3.4.19	4.5.4.19	X	
Personnel heater	3.4.20	4.5.4.20	X	
Heater fuel pressure	3.4.20.1	4.5.4.20	X	
Engine smoke grenade	3.4.21	4.5.4.21	X	
Arming	3.4.21.1.1	4.5.4.21.1.1	X	
Circuit voltage	3.4.21.1.2	4.5.4.21.1.2	X	
Vibration and noise level	3.4.22	4.5.4.22	X	

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TABLE V. One thousand mile test cycle.

Course	Mileage and speed 1/ mi (km)
Hard surface roads	250 (400)
Gravel and dirt roads	250 (400)
Level cross country	250 (400)
Hilly cross country	250 (400)

1/ Speeds shall be varying up to the maximum safe speed in all applicable gear ranges depending on course and conditions.

4.4 Examination. Visual, dimensional, and primary functional examination shall consist of examination of the chassis for conformance to the applicable drawings and this specification (see table III). Sampling shall be in accordance with 4.5.3.

4.5 Methods of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.5.2 Defects. Conformance to 3.2, 3.3 and 3.5 thru 3.10 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

4.5.3 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5.4 Performance. Conditions for tests hereunder shall be as specified in 3.4.

4.5.4.1 Power plant and power train test. To determine conformance to 3.4.1, the chassis shall be subjected to the steering and braking tests (see 4.5.4.6 and 4.5.4.11).

4.5.4.2 Environments test. To determine conformance to 3.4.2, the chassis shall be subjected to the initial production and comparison tests (see 4.2.2.1).

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4.5.4.2.1 Cooling system test. To determine conformance to 3.4.2.1, the chassis shall be subjected to the initial production and comparison tests (see 4.2.2.1).

4.5.4.3 Acceleration test. To determine conformance to 3.4.3, the chassis shall be operated as specified herein and performance timed.

4.5.4.4 Level road speeds test. To determine conformance to 3.4.4, the chassis shall be operated under the specified conditions. A speed of not less than 30 mph (48 km/h) shall be maintained for 10 minutes with engine at full governed speed. A speed of not less than 2.5 mph (4 km/h) shall be maintained 5 minutes with engine at full governed speed and transmission in low gear.

4.5.4.5 Level road drift test. To determine conformance to 3.4.5, chassis shall be operated on smooth, paved surface at speed indicated. Test shall be run twice in each direction and the average drift at end of four runs shall not exceed requirements.

4.5.4.6 Level road stopping test. To determine conformance to 3.4.6, the chassis shall be operated on smooth, paved surface at speed indicated. Stopping distance shall be measured and shall not exceed requirements.

4.5.4.7 Grade and slope tests. To determine conformance to 3.4.7, 3.4.7.1, 3.4.7.2, 3.4.7.3, and 3.4.7.4, chassis shall be operated on the grades and slopes indicated and performance observed.

4.5.4.8 Shallow water fording test. To determine conformance to 3.4.8, chassis shall be operated in 48 in. (1220 mm) of water for not less than the time indicated and water depth in the hull measured.

4.5.4.8.1 Engine starting (shallow water) test. To determine conformance to 3.4.8.1, the chassis and the engine, shall be operated as required and the engine observed for proper restart within indicated time. Accessories and instruments shall be observed for proper operation.

4.5.4.8.2 Lubricant contamination test. To determine conformance to 3.4.8.2, after completion of fording test, lubricant samples shall be tested for water content. Test results shall be submitted to the Government inspector.

4.5.4.9 Trench crossing test. To determine conformance to 3.4.9, chassis shall be operated over obstacles of the dimensions stated and observed for performance and visually examined for physical damage.

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4.5.4.10 Vertical obstacles test. To determine conformance to 3.4.10, chassis shall be operated over obstacles of the dimensions stated, observed for performance, and visually examined for physical damage.

4.5.4.11 Turning test. To determine conformance to 3.4.11, chassis shall be operated as stated. When measured, the pivot-circle shall not exceed requirement.

4.5.4.12 Sealing test. To determine conformance to 3.4.12, seals shall be visually examined after completion of all mobile tests.

4.5.4.12.1 Escape hatch seal test. To determine conformance to 3.4.12.1, chassis shall be exposed to heavy-rain conditions (natural or simulated) and hatch seal visually examined for leakage. Hatch shall also be observed for leakage during fording test (see 4.5.4.8).

4.5.4.13 Controls mechanisms test. To determine conformance to 3.4.13, controls shall be observed for proper function during mobile test operations.

4.5.4.14 Adjustment mechanisms test. To determine conformance to 3.4.14, after mobile test operations, adjustment mechanics shall be observed for adjustment maintenance.

4.5.4.15 Generator and blower tests. To determine conformance to 3.4.15.1 and 3.4.15.2 with engine running at various speeds, generator voltage shall be tested for requirements. Air flow to the generator shall be observed for continuous flow.

4.5.4.16 Air cleaner blower test. To determine conformance to 3.4.16 with engine running, all air outlets shall be examined for continuous air flow. Further examination shall show that air flow stops when engine is switched off.

4.5.4.17 Lights. Conformance to 3.4.17 shall be determined during visual, dimensional, and primary functional examination (see 4.5.2).

4.5.4.18 Locking, unlocking and actuated mechanisms.

4.5.4.18.1 Hull drain valves test. To determine conformance to 3.4.18.1, drain valve levers shall be unlocked and the effort required to open the valves measured. Effort shall not exceed requirement for each valve.

CAUTION: Valves shall be closed and locked after the test.

4.5.4.18.2 Escape hatch release test. To determine conformance to 3.4.18.2, the force required to release hatch locking device shall be measured and shall be within required tolerances. Hatch shall be reclosed and chassis subjected to shallow water fording test (see 4.5.4.8).

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4.5.4.18.3 Operator's seat test. To determine conformance to 3.4.18.3, with an average weight person (180 lbs (80 kg) nominal) in the seat, force required to actuate adjustment lever shall be measured and shall not exceed requirement.

4.5.4.18.4 Fuel shut-off test. To determine conformance to 3.4.18.4, the engine shall be operated at 50% of governed rpm, manual shut-off valve shall be actuated to "off" position. Time required for engine to stop shall be measured and shall not exceed requirement.

CAUTION: Fuel valve shall be returned to "ON" position and the system primed and bled of air.

4.5.4.19 Fire extinguisher. To determine conformance to 3.4.19, the fire extinguisher shall be tested for operation. With the engine idling at 700 to 800 rpm, the internal control shall be pulled with a maximum effort of 55 lbs (25 kg) using a force gauge. The first shot shall actuate one bottle. After a time delay of more than 6 seconds but less than 11 seconds, the engine shall have stopped and the system shall begin discharging CO_2 into the engine compartment and in the center of the engine "V". After returning the handle to its original position, a second actuation shall discharge the remaining two bottles of CO_2. Removal and weighing of each CO_2 bottle after test shall show a minimum of 9 lbs (4 kg) of CO_2 discharged during test.

CAUTION: When conducting these tests, safety precautions shall be exercised (see 6.3.1).

4.5.4.20 Personnel heater test. To determine conformance to 3.4.20 and 3.4.20.1, the heater shall be started and observed for proper operation and tested for conformance to all requirements.

4.5.4.21 Engine smoke generator test. To determine conformance to 3.4.21, activate the engine smoke generator switch when the engine is warm and operating at 1600 rpm. Verify that both right and left exhaust ducts emit white smoke.

NOTE: Do not test the engine smoke generator for more than 15 seconds, nor within a building, nor with personnel exposed to the exhaust plume.

4.5.4.21.1 Smoke grenade discharger circuits tests.

4.5.4.21.1.1 Arming test. To determine conformance to 3.4.21.1.1, operate the master battery switch to ON and the grenade system power switch to ON, and verify that the power-on (ready) lamp in the grenade power box illuminates. Operate the grenade system power switch to OFF and verify that the power-on (ready) lamp is extinguished.

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4.5.4.21.1.2 Circuit voltage test. To determine conformance to 3.4.21.1.2, arm the circuit and verify that the power-on (ready) lamp is illuminated. Actuate the RIGHT firing pushbutton and verify that the voltage on the smoke grenade discharger pins is as specified. Actuate the LEFT firing pushbutton and verify that the voltage on the smoke grenade discharger pins is as specified. Return power-on (ready) switch to OFF and verify that voltage is not present on any of the firing pins.

4.5.4.22 Vibration and noise level test. To determine conformance to 3.4.22, the average of three noise level readings at each octave band shall be determined at each operator's head position under normal operating procedures. Verify that the noise hazard signs are clearly visible to personnel in all crew stations.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The tank chassis is intended for use as a transporting launcher for an AVL bridge after subsequent adaptation of the launching and bridge mechanism to the chassis.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. When first article is required (see 3.1).
d. Painting, if other than specified (see 3.9.1).
e. If preproduction inspection is required (see 4.2.1).
f. If initial production inspection is required (see 4.2.2).
g. Packaging requirements (see 5.1).

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6.3 Safety precautions.

6.3.1 Fire extinguisher. Caution should be exercised in handling carbon dioxide fire extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire extinguisher control system since physical injury is highly probable.

6.4 Subject term (key word) listing.

Components
Mover
Power train

6.5 Leaks. The following definitions should be used for examination of defects for leaks:

a. Weep: - Any evidence of fluid beyond the seal.
b. Seep: - Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplet: - Any evidence of fluid beyond the seal that results in the formation of a droplet.
d. Drip: - Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Break-in run. The break-in run will include preliminary checks of suspension, controls, power plant, etc., and provide wear-in for final adjustments. The break-in run will include the following activities and conditions:

a. Preparation. Prior to break-in run, except for equipment and component installations and servicing performed by the contractor as a final processing of the chassis for acceptance, the chassis should be completely assembled and all adjustments completed. The chassis, including all applicable components requiring lubrication, should be lubricated in accordance with production lubrication chart D 10934403. Engine oil conforming to the applicable seasonal grade of MIL-L-2104 should be used. Specific applications by grades or types for various expected seasonal temperature ranges should be as specified. Fuel/water separator and primary fuel filter should be bled of air by loosening each respective bleed plug and operating the in-tank fuel pump. After air is bled, tighten the bleed plugs.
b. Oil pressure and temperature. Prior to the start of the break-in run, the engine should be operated at 800 to 1200 rpm until lubricating oil is at operating pressure and temperature. If at any time during break-in run, the engine has been stopped for at least 30 minutes, the engine should again be operated as above before continuing break-in run. Proper oil pressure and temperature should be maintained during operation on level ground, 60% grades, and 30% side slopes.

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c. Operation and distance. Each chassis should be given a break-in run for the distances specified in table VI on smooth, level, hardsurfaced roads. The chassis should be operated over the undulating section of road (see figure 1) during "a" and "b" divisions of the break-in run, the undulating section will be bypassed.

TABLE VI. Speeds and distances for break-in run.

Division of run	Speed mph (km/h)	Distance, mi (km)	Test conditions
a	0 to 10 (0 to 16)	10 (16)	Track (road)
b	11 to 15 (17 to 24)	15 (24)	Track (road)
c	16 to 20 (25 to 32)	10 (16)	Track (road)
d	21 to max. (33 to max.)	10 (16)	Track (road)

d. Reverse operation. After each division of the break-in run, the chassis should be stopped, engine allowed to idle for not less than 2 minutes, and then driven in reverse for a distance of not less than 50 ft (15 m) at speeds between 2 and 7 mph (3 and 11 km/h).

e. Condition after run. After completion of the break-in run, the engine should idle between 700 and 750 rpm. Prior to chassis being submitted to the Government for acceptance, all suspension wheel bearings should be checked to determine if adjustments are correct and readjusted as required. There should be no defects indicative of damaged components or maladjustments that may cause faulty chassis operation.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

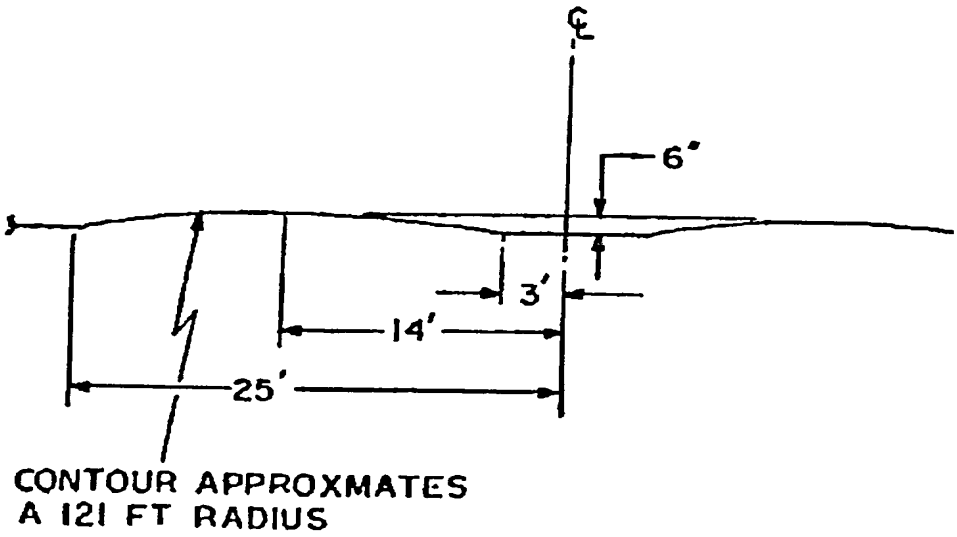


FIGURE 1. Profile of test track (artificial bump).

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Army - AT

Preparing Activity:
Army - AT

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62267A(AT)	2. DOCUMENT DATE (YYMMDD) 970127

3. DOCUMENT TITLE

Chassis, Tank: Armored Vehicle, Bridge Launcher, M60A1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (810) 574-8745 (2) AUTOVON: 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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